

Inner Products.

Definition: Let V be a vector space over $\mathbb{C}$. An inner product on V is a complex-valued function of two ordered vectors in V , denoted by $(\cdot, \cdot)$, which satisfies the following conditions.

- (I) $(\vec{x}, \vec{x}) \geq 0$ for all $\vec{x} \in V$ and $(\vec{x}, \vec{x}) = 0$ if and only if $\vec{x} = \vec{0}$.
- (II) $(\vec{x}, \vec{y}) = \overline{(\vec{y}, \vec{x})}$ for all $\vec{x}, \vec{y} \in V$.
- (III) $(c\vec{x}, \vec{y}) = c(\vec{x}, \vec{y})$ for all $\vec{x}, \vec{y} \in V$ and $c \in \mathbb{C}$.
- (IV) $(\vec{x} + \vec{y}, \vec{z}) = (\vec{x}, \vec{z}) + (\vec{y}, \vec{z})$ for all $\vec{x}, \vec{y}$, and $\vec{z} \in V$.

Remarks: $(\vec{x}, c\vec{y}) = \overline{(c\vec{y}, \vec{x})} = \overline{c(\vec{y}, \vec{x})} = \bar{c} \overline{(\vec{y}, \vec{x})} = \bar{c}(\vec{x}, \vec{y})$

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Definition: Let V be a vector space over $\mathbb{R}$. An inner product on V is a **real**-valued function of two ordered vectors in V , denoted by $(\cdot, \cdot)$, which satisfies the following conditions.

- (I) $(\vec{x}, \vec{x}) \geq 0$ for all $\vec{x} \in V$ and $(\vec{x}, \vec{x}) = 0$ if and only if $\vec{x} = \vec{0}$.
- (II) $(\vec{x}, \vec{y}) = (\vec{y}, \vec{x})$ for all $\vec{x}, \vec{y} \in V$.
- (III) $(c\vec{x}, \vec{y}) = c(\vec{x}, \vec{y})$ for all $\vec{x}, \vec{y} \in V$ and $c \in \mathbb{R}$.
- (IV) $(\vec{x} + \vec{y}, \vec{z}) = (\vec{x}, \vec{z}) + (\vec{y}, \vec{z})$ for all $\vec{x}, \vec{y}$, and $\vec{z} \in V$.

Definition. A vector space V over F endowed with a specific inner product is called an inner product space. If $F = \mathbb{C}$, we call V a complex inner product space. If $F = \mathbb{R}$, we call V a real inner product space.

Examples: 1) Let $V = \mathbb{R}^n$. $\vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$, $\vec{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$ define $(\vec{x}, \vec{y}) = \vec{x}^T \vec{y} = (x_1 \ x_2 \ \dots \ x_n) \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$
 $= x_1 y_1 + x_2 y_2 + \dots + x_n y_n.$

2) Let $V = \mathbb{C}^n$. $\vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$, $\vec{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$ define: $(\vec{x}, \vec{y}) = \vec{y}^* \vec{x} = (x_1 \ x_2 \ \dots \ x_n) \begin{pmatrix} \bar{y}_1 \\ \bar{y}_2 \\ \vdots \\ \bar{y}_n \end{pmatrix}$
 $= x_1 \bar{y}_1 + x_2 \bar{y}_2 + \dots + x_n \bar{y}_n$
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The above inner products on $\mathbb{R}^n$ or $\mathbb{C}^n$ are referred as standard inner product

3) Let $V = \mathbb{R}^{n \times n}$. define $(A, B) = \text{Tr}(A^T B)$.

Check: i). $(A, A) = \text{Tr}(A^T A) \geq 0$ b/c $\text{Tr}(A^T A) = \sum_{j=1}^n \sum_{i=1}^n a_{ij}^2 \geq 0$.

and $(A, A) = 0$ if and only if $A = 0$. (Homework problem result)

II). $(A, B) = \text{Tr}(A^T B) = \text{Tr}(\underbrace{(A^T B)^T}) = \text{Tr}(B^T A) = (B, A)$

b/c Trace of a matrix equals to the trace of its transpose.

III). $(cA, B) = \text{Tr}((cA)^T B) = \text{Tr}(c A^T B) = c \text{Tr}(A^T B) = c(A, B)$

IV) $(A+B, C) = \text{Tr}((A+B)^T C) = \text{Tr}(A^T + B^T)C = \text{Tr}(A^T C + B^T C)$
 $= \text{Tr}(A^T C) + \text{Tr}(B^T C) = (A, C) + (B, C).$

4). Let $V = \mathbb{C}^{n \times n}$. Check that $(A, B) = \text{Tr}(B^* A)$ defines an inner product on $\mathbb{C}^{n \times n}$.

5). Let $V = C^0([-1, 1])$ be the vector space of the set of continuous real-valued functions on $[-1, 1]$. Define

$$(f, g) = \int_{-1}^1 f(x) g(x) dx.$$

Check: i). $(f, f) = \int_{-1}^1 f(x)^2 dx \geq 0$. and $(f, f) = 0$ if and only if $f = 0$.

II) $(f, g) = \int_{-1}^1 f(x) g(x) dx = \int_{-1}^1 g(x) f(x) dx = (g, f)$

III) $(cf, g) = \int_{-1}^1 cf(x) g(x) dx = c \int_{-1}^1 f(x) g(x) dx = c(f, g)$

$$\begin{aligned}
 \text{IV) } (f+g, h) &= \int_1^1 (f(x) + g(x)) h(x) dx = \int_1^1 [f(x)h(x) + g(x)h(x)] dx \\
 &= \int_1^1 f(x)h(x) dx + \int_1^1 g(x)h(x) dx \\
 &= (f, h) + (g, h).
 \end{aligned}$$

Def'n: Let $A \in \mathbb{R}^{n \times n}$. Then A is (Symmetric) positive definite if $A^T = A$ and $\vec{x}^T A \vec{x} > 0$ for all $\vec{x} \neq \vec{0}$.

Def'n: Let $A \in \mathbb{R}^{n \times n}$. Then A is (Symmetric) positive Semidefinite if $A^T = A$ and $\vec{x}^T A \vec{x} \geq 0$ for all $\vec{x}$.

6). Let $A \in \mathbb{R}^{n \times n}$ be symmetric positive definite. $\forall \vec{x}, \vec{y} \in \mathbb{R}^n$, define $(\vec{x}, \vec{y}) = \vec{x}^T A \vec{y}$. Then $(,)$ defines an inner product on $\mathbb{R}^n$.

Def'n: Let $A \in \mathbb{C}^{n \times n}$. Then A is (Hermitian) positive definite if $A^* = A$ and $\vec{x}^* A \vec{x} > 0$ for all $\vec{x} \neq \vec{0}$.

Def'n: Let $A \in \mathbb{C}^{n \times n}$. Then A is (Hermitian) positive semidefinite if $A^* = A$ and $\vec{x}^* A \vec{x} \geq 0$ for all $\vec{x}$.

7). Let $A \in \mathbb{C}^{n \times n}$ be (Hermitian) positive definite. For any $\vec{x}, \vec{y} \in \mathbb{C}^n$ define $(\vec{x}, \vec{y}) = \vec{y}^* A \vec{x}$. Then $(,)$ defines an inner product on $\mathbb{C}^n$.

Proposition. Let V be an inner product space over $F = \mathbb{C}$ or $\mathbb{R}$. Then for any $\vec{x}, \vec{y}$ and $\vec{z} \in V$ and $c \in F$. The following statements are true.

- 1). $(\vec{x}, \vec{y} + \vec{z}) = (\vec{x}, \vec{y}) + (\vec{x}, \vec{z})$
- 2). $(\vec{x}, c\vec{y}) = c(\vec{x}, \vec{y})$
- 3). $(\vec{x}, \vec{0}) = (\vec{0}, \vec{x}) = 0$
- 4). $(\vec{x}, \vec{x}) = 0$ if and only if $\vec{x} = \vec{0}$
- 5). If $(\vec{x}, \vec{y}) = (\vec{x}, \vec{z})$ for all $\vec{x} \in V$, then $\vec{y} = \vec{z}$.

Proof. Exercise.

Definition. Let V be an inner product space. For $\vec{x} \in V$, we define the norm

(or the length) of $\vec{x}$ by $\|\vec{x}\| = \sqrt{(\vec{x}, \vec{x})}$.

Proposition. Let V be an inner product space over F . Then for all $\vec{x}, \vec{y} \in V$ and $c \in F$, the following statements are true.

1) $\|c\vec{x}\| = |c| \|\vec{x}\|$

2) $\|\vec{x}\| = 0$ if and only if $\vec{x} = \vec{0}$. In any case, $\|\vec{x}\| > 0$

3). (Cauchy-Schwarz Inequality) $|(\vec{x}, \vec{y})| \leq \|\vec{x}\| \cdot \|\vec{y}\|$

4). (Triangular Inequality) $\|\vec{x} + \vec{y}\| \leq \|\vec{x}\| + \|\vec{y}\|$.

Proof: 1). $\|c\vec{x}\|^2 = (c\vec{x}, c\vec{x}) = c\bar{c}(\vec{x}, \vec{x}) = |c|^2 \|\vec{x}\|^2$

$\Rightarrow \|c\vec{x}\| = |c| \|\vec{x}\|$.

2). $\|\vec{x}\|^2 = (\vec{x}, \vec{x}) \geq 0$ and $(\vec{x}, \vec{x}) = 0$ iff $\vec{x} = \vec{0}$

Review on complex numbers: If $z = a + bi$ then $\bar{z} = a - bi$, and $|z|^2 = a^2 + b^2 = z\bar{z}$
if $z = \frac{z_1}{z_2}$ then $\bar{z} = \frac{\bar{z}_1}{\bar{z}_2}$.

In particular, if $z = (\vec{x}, \vec{y})$. Then $\bar{z} = (\vec{y}, \vec{x})$ then $z \cdot \bar{z} = (\vec{x}, \vec{y})(\vec{y}, \vec{x}) = |\vec{x}|^2$

3). Case 1: If $\vec{y} = \vec{0}$, $|(\vec{x}, \vec{0})| = |0| = 0 \Rightarrow |(\vec{x}, \vec{y})| = \|\vec{x}\| \cdot \|\vec{y}\|$.
 $\|\vec{x}\| \cdot \|\vec{0}\| = 0 \Rightarrow |(\vec{x}, \vec{y})| \leq \|\vec{x}\| \cdot \|\vec{y}\|$.

Case 2: If $\vec{y} \neq \vec{0}$, $\forall c \in F: 0 \leq \|\vec{x} - c\vec{y}\|^2$

$= (\vec{x} - c\vec{y}, \vec{x} - c\vec{y})$

$= (\vec{x}, \vec{x}) - (\vec{x}, c\vec{y}) - (c\vec{y}, \vec{x}) + (c\vec{y}, c\vec{y})$

$= \|\vec{x}\|^2 - \bar{c}(\vec{x}, \vec{y}) - c(\vec{y}, \vec{x}) + c\bar{c}\|\vec{y}\|^2$

Take $c = \frac{(\vec{x}, \vec{y})}{(\vec{y}, \vec{y})} = \frac{(\vec{x}, \vec{y})}{\|\vec{y}\|^2}$ $\Rightarrow \|\vec{x}\|^2 - \frac{(\vec{y}, \vec{x})}{\|\vec{y}\|^2}(\vec{x}, \vec{y}) - \frac{(\vec{x}, \vec{y})(\vec{y}, \vec{x})}{\|\vec{y}\|^2} + \frac{(\vec{x}, \vec{y})(\vec{y}, \vec{x})}{\|\vec{y}\|^4} \|\vec{y}\|^2$

$$\Rightarrow \bar{c} = \frac{(\vec{y}, \vec{x})}{\|\vec{y}\|^2} = \frac{\|\vec{x}\|^2 - \frac{(\vec{x}, \vec{y})(\vec{y}, \vec{x})}{\|\vec{y}\|^2}}{\|\vec{y}\|^2} = \|\vec{x}\|^2 - \frac{|(\vec{x}, \vec{y})|^2}{\|\vec{y}\|^2}$$

$$\Rightarrow \|\vec{x}\|^2 - \frac{|(\vec{x}, \vec{y})|^2}{\|\vec{y}\|^2} \geq 0 \Rightarrow \|\vec{x}\|^2 \cdot \|\vec{y}\|^2 \geq |(\vec{x}, \vec{y})|^2$$

$$\Leftrightarrow |(\vec{x}, \vec{y})| \leq \|\vec{x}\| \cdot \|\vec{y}\|$$

Review: If $\vec{z} = a + bi$ with $a, b \in \mathbb{R}$. $\operatorname{Re}(\vec{z}) = a$, $\operatorname{Im}(\vec{z}) = b$.

$$\text{Then } \vec{z} + \bar{\vec{z}} = 2a = 2\operatorname{Re}(\vec{z}) \leq 2|\vec{z}|$$

$$\vec{z} - \bar{\vec{z}} = 2bi = 2i\operatorname{Im}(\vec{z}) \leq 2|\vec{z}|$$

$$\begin{aligned} 4). \|\vec{x} + \vec{y}\|^2 &= (\vec{x} + \vec{y}, \vec{x} + \vec{y}) = (\vec{x}, \vec{x}) + (\vec{x}, \vec{y}) + (\vec{y}, \vec{x}) + (\vec{y}, \vec{y}) \\ &= \|\vec{x}\|^2 + 2\operatorname{Re}(\vec{x}, \vec{y}) + \|\vec{y}\|^2 \\ &\leq \|\vec{x}\|^2 + 2|(\vec{x}, \vec{y})| + \|\vec{y}\|^2 \quad \text{by the above review.} \\ &\stackrel{\text{Cauchy-Schwarz}}{\leq} \|\vec{x}\|^2 + 2\|\vec{x}\| \cdot \|\vec{y}\| + \|\vec{y}\|^2 \\ &= (\|\vec{x}\| + \|\vec{y}\|)^2 \end{aligned}$$

$$\Rightarrow \|\vec{x} + \vec{y}\| \leq \|\vec{x}\| + \|\vec{y}\|. \quad \square$$

Orthogonal / orthonormal vectors

Definition. Let V be an inner product space.

1) A set of vectors $\{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_m\} \subseteq V$ is called orthogonal if

$$(\vec{x}_i, \vec{x}_j) = 0 \quad \text{for any } i \neq j, \quad i, j = 1, \dots, m.$$

2). A set of vectors $\{\vec{x}_1, \dots, \vec{x}_m\} \subseteq V$ is called orthonormal if

$$(\vec{x}_i, \vec{x}_j) = \delta_{ij} = \begin{cases} 1, & i=j \\ 0, & i \neq j \end{cases} \quad \text{for all } i, j = 1, \dots, m.$$

Remarks: An orthogonal set may contain the zero vector. An orthonormal set must be all non-zero vectors.

Definition: Let $A \in \mathbb{C}^{n \times n}$. Then A is called a unitary matrix if $A^*A = AA^* = I$

Definition: Let $A \in \mathbb{R}^{n \times n}$. Then A is called an orthogonal matrix if $A^T A = AA^T = I$.

Remarks: 1) A is unitary if and only if all the columns of A form an orthonormal set with respect to the standard inner product on $\mathbb{C}^n$.

2) A is orthogonal if and only if all the columns of A form an orthonormal set with respect to the standard inner product on $\mathbb{R}^n$.

For 1): Denote $A = (\vec{a}_1 \ \vec{a}_2 \ \dots \ \vec{a}_n)$

$$\begin{aligned} \text{Then } A^*A &= \begin{pmatrix} \vec{a}_1^* \\ \vec{a}_2^* \\ \vdots \\ \vec{a}_n^* \end{pmatrix} (\vec{a}_1 \ \vec{a}_2 \ \dots \ \vec{a}_n) \\ &= \begin{pmatrix} \vec{a}_1^* \vec{a}_1 & \vec{a}_1^* \vec{a}_2 & \dots & \vec{a}_1^* \vec{a}_n \\ \vec{a}_2^* \vec{a}_1 & \vec{a}_2^* \vec{a}_2 & \dots & \vec{a}_2^* \vec{a}_n \\ \vdots & \vdots & \ddots & \vdots \\ \vec{a}_n^* \vec{a}_1 & \vec{a}_n^* \vec{a}_2 & \dots & \vec{a}_n^* \vec{a}_n \end{pmatrix} \\ &= \begin{pmatrix} (\vec{a}_1, \vec{a}_1) & (\vec{a}_1, \vec{a}_2) & \dots & (\vec{a}_1, \vec{a}_n) \\ (\vec{a}_2, \vec{a}_1) & (\vec{a}_2, \vec{a}_2) & \dots & (\vec{a}_2, \vec{a}_n) \\ \vdots & \vdots & \ddots & \vdots \\ (\vec{a}_n, \vec{a}_1) & (\vec{a}_n, \vec{a}_2) & \dots & (\vec{a}_n, \vec{a}_n) \end{pmatrix} \end{aligned}$$

$\Rightarrow$ Denote $A^*A = (b_{ij})$ then $b_{ij} = (\vec{a}_j, \vec{a}_i)$

Then $A^*A = I$ iff $(\vec{a}_j, \vec{a}_i) = \begin{cases} 1, & \text{if } i=j \\ 0, & \text{if } i \neq j \end{cases}$

iff $\{\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n\}$ is an orthonormal set. // end of Feb 17